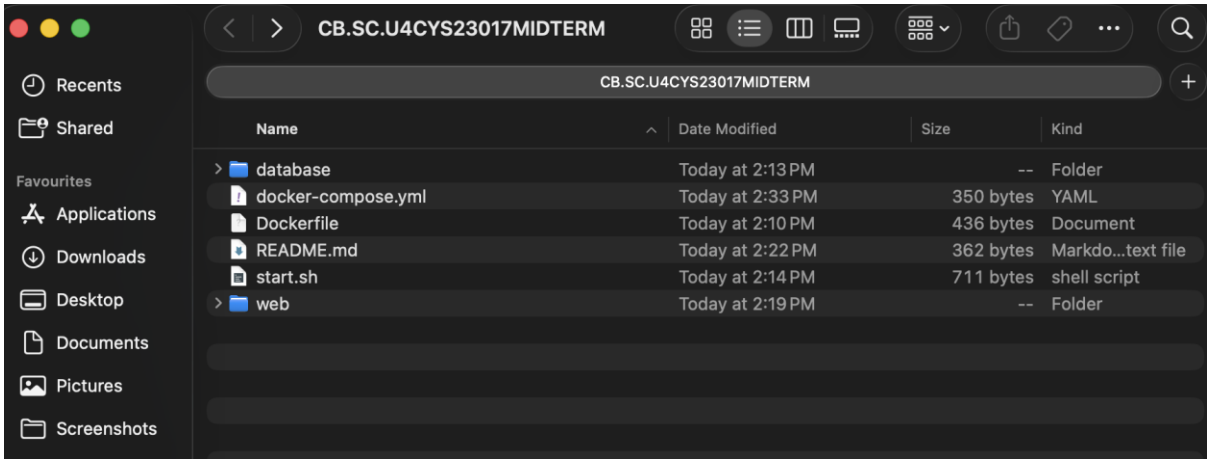
20CYS402 Distributed Systems and Cloud Computing

HIBA LEEYAKATH S

CB.SC.U4CYS23017

MIDTERM EXAM



README.md > # Dockerized Web Authentication Application

```
# Dockerized Web Authentication Application

Roll Number: CB.SC.U4CYS23017
Name: Hiba Leeyakath S
Email: CB.SC.U4CYS23017@cb.amrita.edu

Docker Image:
mt-webauth

Database:
midterm_db

Database User:
midterm_user

Database Password:
midterm_pass

Sample Login:
Username: admin
Password: admin123

Run:
docker compose up -d --build

Access:
http://localhost:5000
```

database > init.sql > ...

```
CREATE TABLE IF NOT EXISTS users (
    id SERIAL PRIMARY KEY,
    username VARCHAR(100) UNIQUE NOT NULL,
    password VARCHAR(100) NOT NULL
);

INSERT INTO users (username, password)
VALUES ('admin', 'admin123')
ON CONFLICT (username) DO NOTHING;

GRANT ALL PRIVILEGES ON TABLE users TO midterm_user;
GRANT USAGE, SELECT ON SEQUENCE users_id_seq TO midterm_user;
```

Welcome | init.sql | README.md | Dockerfile | app.py 2

web > app.py > ...

```python
from flask import Flask, request
import psycopg2
import os

app = Flask(__name__)

@app.route("/", methods=["GET", "POST"])
def login():

    message = ""

    if request.method == "POST":

        username = request.form["username"]
        password = request.form["password"]

        db = psycopg2.connect(
            host="127.0.0.1",
            database="midterm_db",
            user="midterm_user",
            password="midterm_pass"
        )

        cursor = db.cursor()

        cursor.execute(
            "SELECT * FROM users WHERE username=%s AND password=%s",
            (username, password)
        )

        if cursor.fetchone():
            message = "Login Successful! Welcome, " + username
        else:
            message = "Invalid Username or Password!"

        cursor.close()
        db.close()

    return f"""
    <html>
    <head>
        <title>Web Authentication</title>

        <style>
            body {{
                font-family: Arial;
                margin: 40px;
            }}

            h1 {{
                color: #333;
            }}

            input {{
                padding: 5px;
                margin: 5px;
            }}

            button {{
                padding: 5px 12px;
            }}

            .message {{
                margin-top: 25px;
            }}

            .student {{
                margin-top: 25px;
                font-size: 14px;
                color: #555;
            }}
        </style>
    </head>

    <body>

        <h1>Web Authentication</h1>

        <form method="POST">

            Username:
            <input type="text" name="username" required>
            <br>

            Password:
            <input type="password" name="password" required>
            <br>

            <button type="submit">Login</button>

        </form>

        <h3 class="message">{message}</h3>

        <div class="student">
            Name: {os.environ.get("NAME")}<br>
            Roll No: {os.environ.get("ROLLNUMBER")}
        </div>

    </body>
    </html>
    """

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5000)
```

Dockerfile

```
FROM python:3.12-slim

LABEL EMAIL="CB.SC.U4CYS23017@cb.amrita.edu"
LABEL "PRODUCTION VERSION"="1.0"

ENV ROLLNUMBER="CB.SC.U4CYS23017"
ENV NAME="Hiba Leeyakath S"

WORKDIR /midterm-exam

RUN apt-get update && \
    apt-get install -y postgresql postgresql-contrib && \
    rm -rf /var/lib/apt/lists/*

RUN pip install --no-cache-dir flask psycopg2-binary

COPY start.sh /start.sh
RUN chmod +x /start.sh

EXPOSE 5000

CMD ["/start.sh"]
```

Dockerfile

```
FROM python:3.12-slim

LABEL EMAIL="CB.SC.U4CYS23017@cb.amrita.edu"
LABEL "PRODUCTION VERSION"="1.0"

ENV ROLLNUMBER="CB.SC.U4CYS23017"
ENV NAME="Hiba Leeyakath S"

WORKDIR /midterm-exam

RUN apt-get update && \
    apt-get install -y postgresql postgresql-contrib && \
    rm -rf /var/lib/apt/lists/*

RUN pip install --no-cache-dir flask psycopg2-binary

COPY start.sh /start.sh
RUN chmod +x /start.sh

EXPOSE 5000

CMD ["/start.sh"]

```

$ start.sh

```
#!/bin/bash

set -e

service postgresql start

until pg_isready -h 127.0.0.1 -U postgres >/dev/null 2>&1
do
    sleep 1
done

if ! runuser -u postgres -- psql -tAc "SELECT 1 FROM pg_roles WHERE rolname='midterm_user'" | grep -q 1
then
    runuser -u postgres -- psql -c "CREATE USER midterm_user WITH PASSWORD 'midterm_pass';"
fi

if ! runuser -u postgres -- psql -tAc "SELECT 1 FROM pg_database WHERE datname='midterm_db'" | grep -q 1
then
    runuser -u postgres -- createdb -O midterm_user midterm_db
fi

runuser -u postgres -- psql -c "ALTER USER midterm_user WITH PASSWORD 'midterm_pass';"

runuser -u postgres -- psql -d midterm_db -f /midterm-exam/database/init.sql

exec python /midterm-exam/web/app.py

```

docker-compose.yml

```yaml
services:

  webauth:
    build:
      context: .
      dockerfile: Dockerfile

    image: mt-webauth

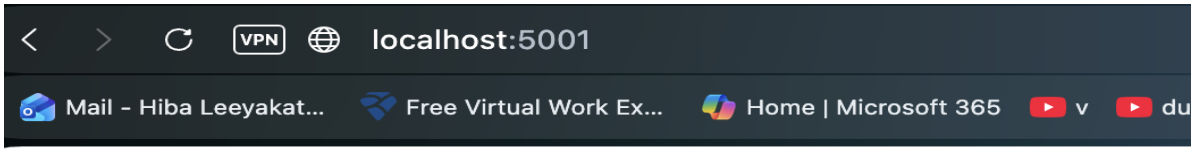
    container_name: mt-webauth-container

    environment:
      ROLLNUMBER: "CB.SC.U4CYS23017"
      NAME: "Hiba Leeyakath S"

    ports:
      - "5001:5000"

    volumes:
      - ./web:/midterm-exam/web
      - ./database:/midterm-exam/database
```

```
hibaleeyakath@Hiba-2 CB.SC.U4CYS23017MIDTERM % docker compose up -d --build
[+] Building 3.8s (13/13) FINISHED
 => [internal] load local bake definitions
 => => reading from stdin 569B
 => [internal] load build definition from Dockerfile
 => => transferring dockerfile: 475B
 => [internal] load metadata for docker.io/library/python:3.12-slim
 => [internal] load .dockerignore
 => => transferring context: 2B
 => [1/6] FROM docker.io/library/python:3.12-slim@sha256:dd29372629eeba2dd003fd9e9d35a5b8236c44727875a0364254b5127af88e65
 => => resolve docker.io/library/python:3.12-slim@sha256:dd29372629eeba2dd003fd9e9d35a5b8236c44727875a0364254b5127af88e65
 => [internal] load build context
 => => transferring context: 30B
 => CACHED [2/6] WORKDIR /midterm-exam
 => CACHED [3/6] RUN apt-get update &&     apt-get install -y postgresql postgresql-contrib &&     rm -rf /var/lib/apt/lists/*
 => CACHED [4/6] RUN pip install --no-cache-dir flask psycopg2-binary
 => CACHED [5/6] COPY start.sh /start.sh
 => CACHED [6/6] RUN chmod +x /start.sh
 => exporting to image
 => => exporting layers
 => => exporting manifest sha256:129d5a5097add960db6ccc20d48822ddfafb9665a6fc67ca1592507aa7de6826
 => => exporting config sha256:f311141eac2ee447f4b612fac525d7152aa21947e511c97d73c79c48fc1ef5ac
 => => exporting attestation manifest sha256:1a22b8d90a8e1eeedfb7d777ae9d2f62d460f7c93a1d54beaeca081aec6e587a
 => => exporting manifest list sha256:8c514f519e01140d801da011b863edcdb683ad2e5eff9cba5ecc0cc0442d2970
 => => naming to docker.io/library/mt-webauth:latest
 => => unpacking to docker.io/library/mt-webauth:latest
 => resolving provenance for metadata file
[+] up 3/3
 ✔ Image mt-webauth                      Built
 ✔ Network cbscu4cys23017midterm_default Created
 ✔ Container mt-webauth-container        Started
```



Web Authentication

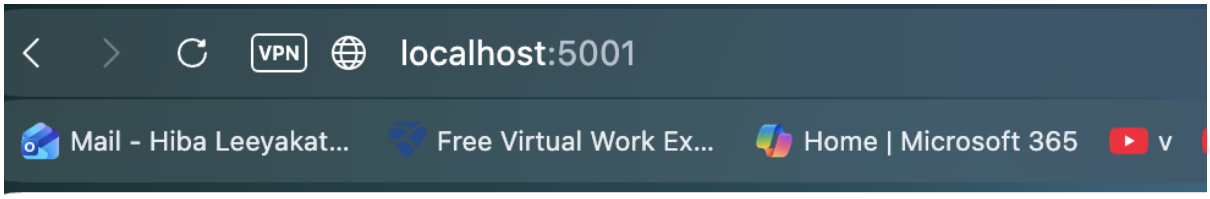
Username: admin

Password: ••••••••

Login

Login Successful! Welcome, admin

Name: Hiba Leeyakath S
Roll No: CB.SC.U4CYS23017



Web Authentication

Username: hiba

Password: •••••••

Login

Invalid Username or Password!

Name: Hiba Leeyakath S
Roll No: CB.SC.U4CYS23017

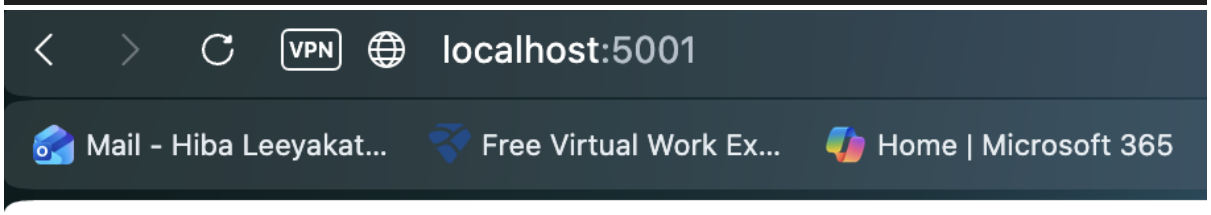
```
[hibaleeyakath@Hiba-2 CB.SC.U4CYS23017MIDTERM % docker exec mt-webauth-container printenv ROLLNUMBER
CB.SC.U4CYS23017
[hibaleeyakath@Hiba-2 CB.SC.U4CYS23017MIDTERM % docker exec mt-webauth-container printenv NAME
Hiba Leeyakath S
```

```
[hibaleeyakath@Hiba-2 CB.SC.U4CYS23017MIDTERM % docker image inspect mt-webauth
[
    {
        "Id": "sha256:bb64bc1b965ff5d0e0d4a7b663fbfd5cd6c389dae4437ec760b54bef826a902b",
        "RepoTags": [
            "mt-webauth:latest"
        ],
        "RepoDigests": [
            "mt-webauth@sha256:bb64bc1b965ff5d0e0d4a7b663fbfd5cd6c389dae4437ec760b54bef826a902b"
        ],
        "Comment": "buildkit.dockerfile.v0",
        "Created": "2026-08-14T08:56:25.174786845Z",
        "Config": {
            "ExposedPorts": {
                "5000/tcp": {}
            },
            "Env": [
                "PATH=/usr/local/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin",
                "LANG=C.UTF-8",
                "GPG_KEY=7169605F62C751356D054A26A821E680E5FA6305",
                "PYTHON_VERSION=3.12.14",
                "PYTHON_SHA256=5c8462af5790baf43a321a1559dbe0db06d1be4300fb85fb53c40060668e548a",
                "ROLLNUMBER=CB.SC.U4CYS23017",
                "NAME=Hiba Leeyakath S"
            ],
            "Cmd": [
                "/start.sh"
            ],
            "WorkingDir": "/midterm-exam",
            "Labels": {
                "EMAIL": "CB.SC.U4CYS23017@cb.amrita.edu",
                "PRODUCTION VERSION": "1.0",
                "com.docker.compose.project": "cbscu4cys23017midterm",
                "com.docker.compose.service": "webauth",
                "com.docker.compose.version": "5.1.4"
            },
            "ArgsEscaped": true
        },
        "Architecture": "arm64",
        "Os": "linux",
        "Size": 184216490,
        "RootFS": {
            "Type": "layers",
            "Layers": [
                "sha256:c01c35a040a25a51cd473910e3212a46d85fb700a6467c687f231d7edd47cbc1",
                "sha256:9bbd22575d4204fc6abbf4a42991d353534d8e60622ef8ad294ba4e630e462ac",
                "sha256:0778dda6f39e1204b0585aa87377671ca10801bf7bd54b0e71f1c63f807d5829",
                "sha256:0798056c7754203b1f11edb1ee80e9a893bf6965a449318503abb5971fa928ad",
                "sha256:f45c17d9abf17d7b73619e46d7a4998b9cd0c3ba95675a897ed90a1565e10b5c",
                "sha256:34ff95679495e601b45a8355ce4b751d6a3424b3c24201251b86ddbe87dbe6db",
                "sha256:c74d9667a6d1b0afdc236e391328ea1976c55d16de75d7d5e9a29148537a1236",
                "sha256:79d3127e88d64e7c1332d574b4de338cce602f798d44efcde524163f1e39a211",
                "sha256:5f70bf18a086007016e948b04aed3b82103a36bea41755b6cddfaf10ace3c6ef"
            ]
        },
        "Metadata": {
            "LastTagTime": "2026-08-14T09:42:12.819457674Z"
        },
        "Descriptor": {
            "mediaType": "application/vnd.oci.image.index.v1+json",
            "digest": "sha256:bb64bc1b965ff5d0e0d4a7b663fbfd5cd6c389dae4437ec760b54bef826a902b",
            "size": 856
        },
        "Identity": {
            "Build": [
                {
                    "Ref": "pl9itbe654nat4hgmwh01igrh",
                    "CreatedAt": "2026-08-14T09:42:12.827302716Z"
                }
            ]
        }
    }
]
```

```
hibaleeyakath@Hiba-2 CB.SC.U4CYS23017MIDTERM % docker tag mt-webauth hibaleeyakath/mt-webauth:latest
hibaleeyakath@Hiba-2 CB.SC.U4CYS23017MIDTERM % docker push hibaleeyakath/mt-webauth:latest
The push refers to repository [docker.io/hibaleeyakath/mt-webauth]
1b7200988f19: Pushed
4f4fb700ef54: Pushed
7b64630a1753: Pushed
1f3174b73705: Pushed
45eff44aea73: Pushed
941eb3454be9: Pushed
c0680a512a88: Pushed
2544aa22a9e0: Pushed
eb7b04a6c654: Pushed
907658cec655: Pushed
latest: digest: sha256:1055393e139d62c1273dfa0386d3d08cc5aa3a9eb424b622d5358dc667f5ac60 size: 856
```

```
hibaleeyakath@Hiba-2 CB.SC.U4CYS23017MIDTERM % docker pull hibaleeyakath/mt-webauth:latest
latest: Pulling from hibaleeyakath/mt-webauth
Digest: sha256:d23e68d65f8f8e865b10e48c78a4cb525f72ff783a6f5ce4038f8cdd140d5127
Status: Image is up to date for hibaleeyakath/mt-webauth:latest
docker.io/hibaleeyakath/mt-webauth:latest

What's next:
    View a summary of image vulnerabilities and recommendations → docker scout quickview hibaleeyakath/mt-webauth:latest
hibaleeyakath@Hiba-2 CB.SC.U4CYS23017MIDTERM % docker run -d --name mt-webauth -p 5001:5000 hibaleeyakath/mt-webauth:latest
c013a7604665373935d16492b1c3fc55e2fa85d10bec995f800c073cf12231a3
hibaleeyakath@Hiba-2 CB.SC.U4CYS23017MIDTERM % docker ps
CONTAINER ID   IMAGE                          COMMAND      CREATED         STATUS         PORTS                                       NAMES
c013a7604665   hibaleeyakath/mt-webauth:latest   "/start.sh"   6 seconds ago   Up 5 seconds   0.0.0.0:5001->5000/tcp, [::]:5001->5000/tcp   mt-webauth
hibaleeyakath@Hiba-2 CB.SC.U4CYS23017MIDTERM %
```



Web Authentication

Username:

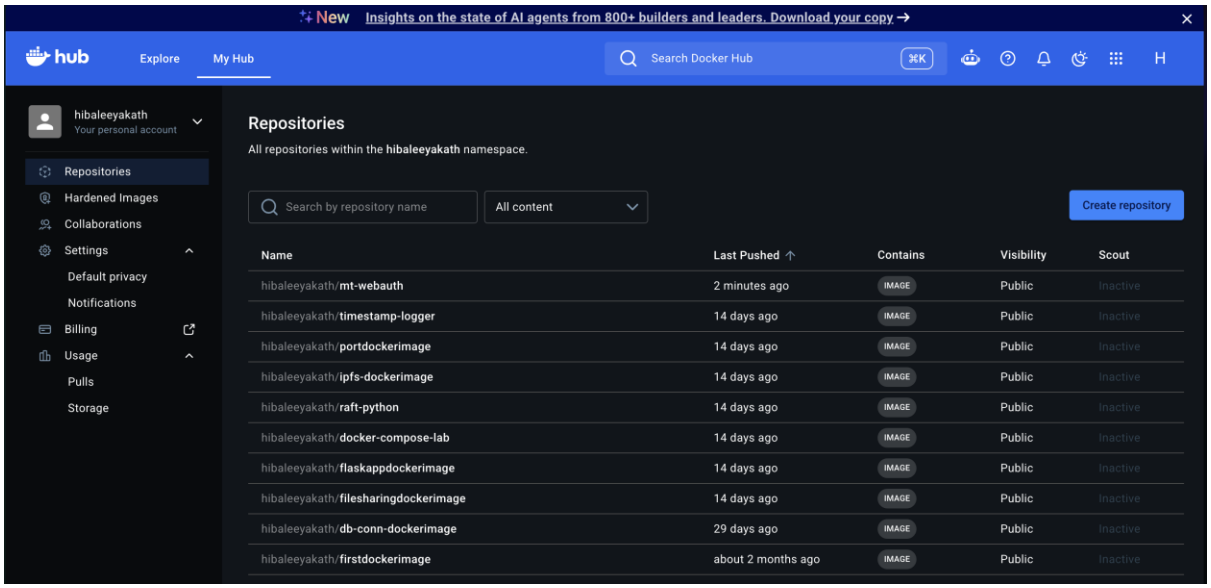
Password:

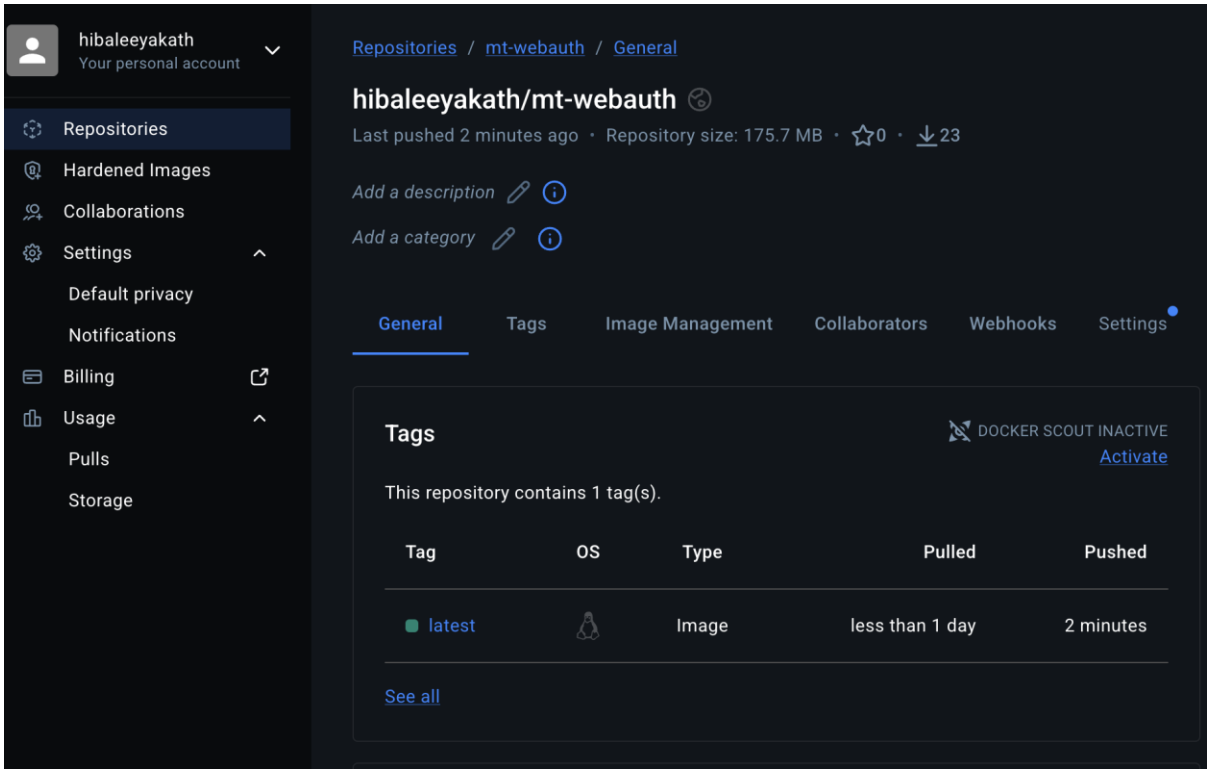
Login

Name: Hiba Leeyakath S
Roll No: CB.SC.U4CYS23017

```
[hibaleeyakath@Hiba-2 CB.SC.U4CYS23017MIDTERM % docker images

IMAGE                                                ID             DISK USAGE   CONTENT SIZE   EXTRA
filesharingdockerimage:1.0                           9136b32fa2d4       285MB         70.7MB
filesharingdockerimage:2.0                           1a189efac7a2       294MB         74.8MB
filesharingdockerimage:latest                        d20db3192385       294MB         74.8MB
firstdockerimage:latest                              6efc16673549       285MB         70.7MB
flaskappdockerimage:latest                           a0e4c9e18373       285MB         70.7MB
hello-world:latest                                   c3cbe1cc1aa5      22.6kB         10.3kB
hibaleeyakath/db-conn-dockerimage:latest             84d98e7b979b       246MB         54.6MB
hibaleeyakath/docker-compose-lab:latest              6b4dc8cb75fc       246MB         86.7MB
hibaleeyakath/filesharingdockerimage:1.0             e5ab7b71f8d3       285MB         70.7MB
hibaleeyakath/filesharingdockerimage:2.0             1a189efac7a2       294MB         74.8MB
hibaleeyakath/filesharingdockerimage:latest          02eefc58b84f       368MB          149MB
hibaleeyakath/firstdockerimage:latest                e26cc243cf70      6.84MB         2.23MB
hibaleeyakath/flaskappdockerimage:latest             37c60f992102       356MB          141MB
hibaleeyakath/ipfs-dockerimage:latest                c45e1cef7f35       292MB          133MB     U
hibaleeyakath/mt-webauth:latest                      d23e68d65f8f       796MB          184MB     U
hibaleeyakath/portdockerimage:latest                 b0af3acc69cd       118MB           52MB
hibaleeyakath/raft-python:latest                     cc950ff99075       277MB         98.7MB
hibaleeyakath/timestamp-logger:latest                ac9eb6126a69       246MB         86.7MB
ipfs/kubo:latest                                     8907cb0cc1ad       133MB         41.8MB
mt-webauth:latest                                    d23e68d65f8f       796MB          184MB     U
nginx:latest                                         ec4ed8b5299e       259MB         64.3MB     U
portdockerimage:latest                               9e0b644696cc        92MB           26MB
portimage:latest                                     523ea931a9be      91.7MB         25.9MB
postgres:latest                                      c2d42a104eb6       671MB          167MB
raft-python-node1:latest                             68a2dd9b82aa       228MB         49.5MB     U
raft-python-node2:latest                             c9349549c3f0       228MB         49.5MB     U
raft-python-node3:latest                             acff85474833       228MB         49.5MB     U
timestamp-logger:latest                              65ff8fe8915e       202MB         43.5MB
```




hibaleeyakath
Your personal account
Repositories
Hardened Images
Collaborations
Settings
Default privacy
Notifications
Billing
Usage
Pulls
Storage
Repositories / mt-webauth / General
hibaleeyakath/mt-webauth
Last pushed 2 minutes ago · Repository size: 175.7 MB · 0 · 23
Add a description
Add a category
General
Tags
Image Management
Collaborators
Webhooks
Settings
Tags
DOCKER SCOUT INACTIVE
Activate
This repository contains 1 tag(s).
Tag
OS
Type
Pulled
Pushed
latest
Image
less than 1 day
2 minutes
See all